# ActInf Livestream #003.1 "A World Unto Itself: Human Communication as Active Inference"

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live streaming is on okay it looks like we are a lot so welcome everybody to team com podcast number three we're really looking forward to this discussion and we really appreciate everyone who has come out to come speak with us so without further ado i'm going to share my screen and we're going to jump into our podcast section so this is team com podcast number three welcome to team com everyone this is a recorded and an archived live stream that we're performing we are an experiment in online team communication and learning and we are the active inference stream for now so people can connect with us via twitter at inferenceactive at this gmail address here we have a public key based team and also if you're watching this you probably know but we're on youtube at this username we really welcome all skills backgrounds and perspectives so please let us know how we can help you and help help provide an on-ramp for wherever you're coming from and please be respectful of the rules of online communication and dialogue so you if you're not speaking if there's noise in the background you can use the raise your hand feature which is on the bottom left of jitsi and so on and please provide us with feedback so we can be improving our work here's going to be the structure of the discussion today first we're going to have a check-in and a warm-up then we're going to briefly cover tin bergens for questions which are a big idea that help us organize some of the concepts that are discussed in the paper then we will get to discussing the paper we'll talk about the goals of the paper and the abstract we'll go through the road map of the paper and talk about the section headers and how the authors get from a to z we'll talk about each one of the figures and try to understand them and then we'll have time for questions throughout as well as at the end so in the check-in or warm-up everyone absolutely feel free to introduce yourself briefly and your location however you'd like and maybe for each of these questions if one or two people could just give a first thought so first warm-up question is what is communication people can raise their hand or just kind of pop corn in yeah go ahead jared uh yeah so alright yeah so hi everyone my name is uh jared faisal i'm uh uh in north carolina in uh in the united states of america um yes uh for me communication is just uh a kind of a set of behaviors and inferences that people use to align their mental states it's kind of like an abstract i think kind of way to think about it but um it inherits from my advisor's line of thought that communication is a sort of tool for people aligning their mental states to a degree good enough for their purposes um yeah that's that's it i'd like to think of it as a behavior that's extended that uh is like a sort of continuous extension of uh behaviors that species across the animal kingdom use um trying to think of it in a way that's anything too too special in linguistics for instance thought of something very very special and unique and it is in a certain way but uh the way that i found it useful to think about communication is as a another sort of behavior in people's toolkits any other thoughts on that yep stefan yeah i would say that i agree with what was just said there actually and uh if you take like an active embodied approach you could say that the um it's an extension communication is an extension of the organism into the environment but one which can happen with interactions with other organisms also extending into their environment and their niche cool and that kind of presages this second question which is what is special interesting or important about human communication and could be nothing could be everything um any thoughts on that one yeah so i have a thought that it's maybe not super special anything about human communication um my name is shannon i'm at the university of california in merced but because of the pandemic i'm at home in south dakota for the foreseeable future um but yeah i think what is special about human communication is that it's the point of view that we're coming from so when we're like studying communication across the animal kingdom it's always the point of comparison either implicitly or explicitly um but you can look at like anything from non-verbal communicative behaviors of like our primate relatives or even of our pets with us um like outside of their species or even of like chemical signals and plants um so maybe there's just a question of how much of something that we want to call agency or intentionality is necessary to call communication important in a specific um like niche or specific animal or plant kingdom or something like that very nice and science as a special form of human communication it's the one that we have if nothing else and so as we kind of continue to build towards the paper how do prior beliefs so beliefs that communicators have about identity narrative or culture play into communication in human or non-human systems sasha um yeah just building on these really great previous answers i'm sasha i'm in davis california um but i just wanted to say that communication is highly culture and context dependent and in thinking about how to align different peoples or organisms uh kind views of the world culture and context plays a big role in deciding on how much to align and whether or not that's important at all cool jared uh yeah so um i wanted to briefly say something about the second point as well i for me real quick the something that's maybe the the main point of interest for me in terms of human communication specifically human communication is the the uh goals of communicators in the human case tend to be or seem to be based on um a lot of for instance developmental and cooperative research um seem to be unique in that the goal is the alignment of mental states per se um so that turns on gathering evidence for that in terms of our interaction so i have to get certain cues from you that you understand what i'm saying and so it's not you know if you grab the thing for me that i i'm asking you to get by accident say accidentally you get the thing that i wanted you to get but you're also giving me cues that you misunderstood what i said and so it was just you know fortuitous circumstance that you happen to get what i wanted even in like young kids you'll see them try to correct that and say no you got the right thing this is really what i wanted and so those sorts that sort of result for instance um suggests that something that's unique is the alignmental states per se um rather than just having some functional outcome which is often the case in for instance other non-human primates forms of communication um and then that turns into prior beliefs for instance how does that play an important role in communication for me that's building what the uh previous speakers said um uh your prior beliefs are the way or an important part of your prior beliefs are the set of cultural conventions that you learn um i.e how a phrase or a a mental state some how a a mental state or intention should be uh made manifest in the environment for others to pick up on that's a set of prior beliefs how do i best do this something that your culture installs in you through all sorts of joint attentional practices earlier in uh earlier and earlier in in development and that's also something that species unique that other species don't don't seem to engage in um is the installation of these kinds of culturally acquired uh patterns and and practices for communicating one's mental states in a way that everyone deems good enough for um sufficient for current purposes i guess cool yeah our last warm-up question is for you what might a satisfying formal framework for communication look like and that might include human and non-human it might include all these features that people are talking about but what would be the attributes of a comprehensive or a formal framework for thinking about all this diversity of communication styles in humans and non-humans what features would that kind of a model be expected or desired to have yes stiffen i think one of the key things is it would have a spatial component in terms of how the dynamics happened in space and around space particularly personal space around the organism and one of the key things about humans is that we can also enfold narrative within itself in story we can and we spatially arrange things so we often say like that's right what's left what's ahead what's been so that ability to somehow tag or embed within the niche is key and then the actual language part is somehow tied to that rather than the other way around which is what a lot of the kind mind-based approaches it tends to be cool shannon i think i'm touching up on that so having this account of how we use space in conversation it also needs to be flexible enough to account what sasha was talking about the way different cultures use space like putting the past in front of you which is awkward to me but makes a lot of sense to other other cultures um and also aspects of communication that aren't linguistic so non-verbal would be obvious but communication and like a musical situation um where no not necessarily any semantic referent is being called upon but all the same we know that we're feeling the same beat or like we're in the same abstract space of interacting with each other cool um yvonne or alex anything to add on any of these points not by no cool i agree about that um aspect where linguistics are one of the modes of communication and yet there's this spatial element and sometimes they blur when you say do you grasp it or can you see what i'm talking about it's very cross-sensory and you can type out what you mean to say yet we're sometimes talking about specific things sometimes not so let's just talk really briefly about tin bergen's four questions which are modeled off of aristotle's four wise and this is a image from cap heim's paper from the u social insect literature and tim bergen's four questions are kind of like a two by two matrix the two aspects on the left side the y axis are whether the domain of analysis is contemporary happening right now or historical in the past and then proximate or ultimate are referring not to ultimate like the best but rather like close and far again kind of a spatial metaphor and the four domains though they sometimes go by slightly different names are causation which is happening approximately and now uh utility which is happening it's the ultimate y it's the it's the teleos and it's happening now as well and then in the proximate version of history so recent history we have development or ontogeny and then the ultimate why of history why do birds have wings why are we here that's captured by evolution and just to put a little bit more of a specific vision on this as we turn towards some of the figures in the paper which will involve birds this is from bateson and lalon 2013 which was a 50-year update on tim bergen's original 63 paper and here's the same four categories of mechanism which is causation uh current utility development and evolution reflected by uh respectively the neuroscience the actual meaning of the song the developmental basis of the song which is showing the songs trace change through time and then the evolutionary history of the trait reflected on a phylogeny so any points on that or can we jump into the paper cool this was just to kind of bring up this idea for those who are maybe coming from outside of a philosophy of biology background because it's a really awesome framework and it helps us get on board with this idea that there's going to be more than one answer to why not just because of just uh different opinions there's actually different kinds of whys when we're talking about why is a bird singing there might be multiple answers just fundamentally so the paper that we are here to talk about today and lucky enough to have jared the first author here for a little bit longer is a world unto itself human communication as active inference and what i sketched the two goals of the paper down as were to formalize cooperative communication as a process of active inference under the free energy principle and to link the mechanistic developmental and evolutionary perspectives on by using active inference as a process theory jared what would you say about that or how does that reflect on what you've intended when you wrote the paper yeah no no i think that's i think that's great and um i want to put particular emphasis within the context of the cs pacific paper on the second point um i really was hoping to um yeah hone in on the idea that all these different scales of analysis are in some sense linked to one another um i.e like you had mentioned in the last slide you know if you want to explain any kind of behavior or inferential pattern or whatever that you see any aspect of a phenotype in in the animal kingdom that's gonna turn in every single case on to fully explain it on on explaining what's going on in each of these different scales of analysis and showing how they all kind of fit into a coherent poll in this case if you take the active inference framework the kind of the idea of gathering evidence for your model your evolutionarily uh favored model um that bottoms out in in real time uh sorts of behavioral patterns um that yeah i was basically trying to link these different scales um and show how they're all sort of part of this evidence gathering procedure um it emerges in this idea of cooperative communication which is something from my advisor mike tomasulo and i hope that uh that this will lead to some sort of uh future more uh technical explicit uh formal analysis cool and when i see tim burgin's four questions i think about yes and because not only is that a tenet improv and communication as we know it at the mechanism time scale to foreshadow but it's like yes there's evolution that's an explanation and there's other kinds of explanations that we want to be addressing as well so it's kind of communication through and through and integration through and through any other comments on the goals the paper before we get to the abstract okay so let's work through the abstract and see how the authors reflected or summarized for everyone what they did they wrote the purpose of this paper is to extend the account of human cooperative by proposing an integrative model of biobehavioral dynamics of cooperative our formulation is based upon active active inference suggests that action perception cycles operate to minimize uncertainty and optimize an individual's internal model of the world we propose that humans are characterized by an evolved adaptive prior belief that their mental states are aligned with or similar to those of con specifics for example that we are the same sort of creature inhabiting the same sort of niche any thoughts or comments on these first parts of the abstract so we're within this integrative framework we want to bring together tin bergen as well as active inference with hopefully an eye towards more formal integrations that are possible in the future the use of cooperative communication emerges as the principal means to gather evidence for this belief again that we're in a shared niche and that we're similar kinds of interactors or english speakers or we're interested in this topic allowing for the development of a shared narrative that is used to disambiguate interactants hidden and inferred mental states any thoughts on that part what does it mean to disambiguate the hidden and inferred mental states well one thought on that shannon go ahead oh yeah um so i guess it's not just to know what the other person might be thinking um or could say in words but also to know that you're both maybe attending to the same reference or that you're both experiencing a similar emotion or if it's different like why is the other person sad right now um so any sort of experience i suppose that two people could be feeling it's to figure out what they're feeling and what you are feeling and yeah a lot of them yeah and we have such a rich internal experience that it's not enough just to infer the external states of a person oh they're sitting down uh big surprise we need to actually do a lot of work to disambiguate or to reduce our uncertainty about hidden states which are mental in this case thus by using cooperative communication individuals effectively attuned to a hermeneutic niche composed in part of others mental states and reciprocally attuned the niche to their own ends via epistemic niche construction uh jared what is a hermeneutic niche and an epistemic niche how are those similar ideas or different yeah so the idea of a hermeneutic niche is that you know when we're when we're communicating we both have our sort of individual perspective so it's you know um it's a an i or a me and a you in addition to that there's you know the we the it's our shared dialogue um we are talking about something um at the same time of course we can decompose that into like i just said our own individual perspectives and so what i was trying to get out there um was the idea of hermeneutics is if you all have read for instance the first and fifth 2015 duet for one paper um they talk about this hermeneutic idea where it comes from the idea of like a deep reading of a text so if you want to really fully understand a text for instance you have to both take words in their immediate context in a sentence in which they're composed then also relate them to the larger context of the paragraph or the the um the chapter or the whole book or whatever um so the idea is relating these two or more different scales of analysis to one another um and seeing it as a sort of integrative explanatory whole um and the epistemic niche construction component uh trying to link up with uh some of the stuff that axel constant um has been doing on niche construction so um we construct our niche so as to offer up very precise uh certain affordances for us i.e we make our niche more easily understandable that could in this case in the instance communication across long stretches of intergenerational time you see these patterns of what using this this terminology epistemic niche construction where um languages cultures build up i should say cultures build up a set of constructions a set of four meaning pairings um [Music] that if you align to that set of four meaning pairings and ontogeny makes your niche very easily i.e easily inferable you can easily grab a hold of others hidden states as there were others mental by using these sorts of uh uh very reliable um sort of epidemic tools or devices which are we talked about in the paper uh communicative constructions and and the way that you describe that it makes me think about the royal we in english and if somebody said well what am i talking about it's almost like wait what are you talking about if you're not sure what you're talking about what are you talking about but when somebody says what are we talking about here it's a rhetorical question that helps us have a shared informational niche and i agree axel's work in that respect is excellent this means that niche construction enables features of the niche to encode precise reliable cues about the deontic or shared value of certain action policies e.g the utility of using communicative constructions to disambiguate mental given expectations about shared prior beliefs what is deontic value in this context jared or anyone else maybe jared first and anyone else can raise their hand i think if if stefan wants to go oh have in a in your environment so um it can be useful so it's kind of a longer standing queue that's kind of established in the niche um just one thing i was thinking when i was seeing this i think it's quite important that they're talking about cooperative communication here um and that narrows it down then to something a bit more instrumental um it's not necessarily coordinated communication so it's not necessarily as action-oriented as that but there is a sense that this is trying to get to we're trying to achieve or trying to cooperate towards something therefore i want to align my mental states because if you're not cooperating with someone you could be just trying to work out the hidden states of like what is it that it's important for me to talk to them about which may not be their mental state it just may be their state in the world at the moment if that makes sense which i think this is like you've already got some alignment and assumed i think right and if it's not cooperative it might be adversarial which kind of takes us to dennett's intentional stance where you just say whatever it is if it's a computer or if it's a human i'm just trying to win this little local game theory interaction and the paper doesn't go that direction it moves towards cooperative communication so jared any thoughts there yeah yeah so this is all great discussion and this um building on the point you just said about you know cooperative versus adversarial sorts of for instance communication um even in the case of you know i say you know i'm using communication or shared language to try to outsmart you or outdo you or something um exercising like a kind of like machiavellian intelligence i think it's been called um that still relies on you know the shared presupposition i guess you want to use that language of um a cooperative bedrock so you know the idea is if i'm using this piece of language typically it's for you know honest genuine communicative goals and the idea for instance deception relies on me using you know kind of exploiting that shared assumption to my own benefit to your detriment um so even at least in the case of humans when you're you know lying or something um adversarial communication or whatever still relies on there being some shared set of cooperative presumptions and that that that is what is exploited um for humans when they're communicating um and also about the the deontic value that those sorts of ideas uh deontic or shared value that um the idea there is that we you know when in ontogeny when you are attuning to learning your culture set of what what these constructions are are in speaker's brains there's um lots of good work for instance from someone who i like joan bybee that suggests that these are very automatized automatic sorts of mappings between form and meaning that at least in their normal use don't require expensive heavy sorts of computations maybe when you're first learning them they do but the more you use them and exercise them in actual joint practice with others uh in aligning mental states they become very automatic and so the idea of geontic value is that it bypasses uh the deontic value of policy bypasses all the kinds of heavy computational updates that might be required by for instance someone learning how to use constructions but yeah and instead provides a very automatic mapping between states so it depends how you chunk a construction but constructions become linked up in a very automatized way and then it can be used effortlessly by speakers um that's the idea of at least in this case the deontical shared value of certain action policies ie constructions cool and and using the niche info niche uh idea it's kind of like a tragedy of the commons if people pollute the environmental it doesn't work as well as it could and if people pollute the informational with deceptive signaling it also doesn't work so it's a the niche metaphor really works well so let's think about the last part of the abstract in turn the alignment of mental states which are prior beliefs enables the emergence of a novel contextualizing scale of cultural dynamics that encompasses the actions and mental states of the ensemble of interactants and their shared environments the dynamics of this contextualizing layer of cultural organizational feedback across gales acts to constrain the variability of prior expectations of the individuals who constitute it and then just to close it with the contribution our theory additionally builds upon the active inference literature by introducing a new set of neurobiologically plausible computational hypotheses for cooperative communication we conclude with directions for future research any last thoughts on the abstract or can we go to the roadmap cool really awesome and it just shows how much there is to unpack in the very distilled representation of the abstract so and jared i know that you have to head out at some point so feel free to just anyone can bounce when they need to yeah i was gonna say i'm gonna i'm gonna head out right now actually i have to go but thanks this was this was wonderful thank you for uh putting us on thank you we would love to have you at a future conversation too so see you later thanks cool so the road map of the paper now now that he's gone and we can really talk about the paper uh it starts with an introduction part two is a theoretical background and they cover several of the key areas of background which are the evolutionary origins and the developmental origins of cooperative communication so two different time scales evo and devo often called then they take stock and table one summarizes some of the key features that are related to cooperative communication so two a b and c don't have to do with free energy they don't have to do it with active inference this is more classical understandings of cooperative communication section three is explicitly about active inference first they talk about the idea of active inference broadly and about how adaptive priors help organisms relate in the world and about the idea of alignment of priors within the active inference framework figure 1 and figure 2 show active inference in the world as well as active inference and communication in the world and we're going to look through each one of the figures one by one they then turn to the topic of deontic values which are shared expectations about the value of policies so section three takes us from what is and why is it important for communication to what is this concept of deontic value in the context of the active inference framework then they bring it together in section four where they uh more clearly talk about human communication as active inference first they look at the dynamics of communication at the time scale of mechanism which is the individual in context as well as the dyad which is the minimum communicative unit then the dynamics of communication at the time scale of ontogeny or development and then zooming back one more level what are the dynamics of communication at the time scale of cultural evolution figure 3 is a really nice representation of the communication and coupled action perception cycle figure 4 has some examples in the bird song as well as figure five of free energy minimization in the case of improvisational duets and figure five is a duet for one learning to synchronize through active inference they then conclude with future directions and conclusions so we're going to get from introductions to communication from an evo devo perspective through the free energy principle inspired framework of active inference to the point where we're actually making useful claims about the world within what we've put together in sections two three four and then leading to future directions what might be interesting for people to study any thoughts on the roadmap or can we go to the table and figures cool so table one would anyone else like to give this a first summary i put up on the top right the remember the cap time figure just so that we remember that these are not uh they're not conflicting uh layers of analysis or scales of analysis they're they're all part of one integrated punnett square if you will and what table one does is and anyone can just raise their hand if they want to jump in but otherwise i'll just kind of keep the error alive table one summarizes some of these key features that have to do with cooperative communication using the terminology in many cases from the non-active inference community so that's why this paper is such a bridge from a conceptual and even a keyword and these four scales of analysis which are reflected by tin bergen's four whys are the real-time and the characteristic dynamics there have to do with ostension which means to show or clearly demarcated joint attention which is people looking at the same thing and in the human case often the the um whites of the eyes surrounding the colors part of the eye have been discussed maybe shannon if you're familiar with uh with the non-human primate literature i know that the non-human primate and humans have quite different eyes and people have speculated that has to do with attention um also at the real time level there's the optimization of relevance which is if someone says okay i i know this part of trigonometry you can skip through that it's not relevant to me or that word wasn't relevant to me because i don't know what it is so could you could you optimize the relevance of your sentence by defining the word coupled bidirectional information flow as well as approximate motivation to align and coordinate mental states uh like if somebody needs something and you want to make sure you have the other person's attention hey i'm driving and i'm thirsty can you get me this water bottle if the other person's not paying attention then you're gonna stay thirsty but if the other person is paying attention it's the motivation that you have to align your mental states so they can get you the water bottle um and there's other characteristic dynamics at the time scale of development phylogeny as well as adaptation any any remarks on this um yeah i was gonna say it makes a lot more sense that this is jared's advisor to understand better the ontogeny of communication or cooperative so i found this an interesting um and really relevant bridge to understanding the onset of cooperative communication in a person's life as well as comparing it across species but yeah i just thought that was a very interesting and irrelevant uh connection to think about how we learn to uh cooperate and make our uh internal states known and that we actually are unsatisfied until we are able to make our internal states known cool stefan yeah i kind of think i always find these these these the way they link with each other kind of interesting um and kind of um could be read different ways um because you could you could the real-time mechanism could stretch out over time in terms of how you think your cooperation is going to have impact on your life um as an as an elder or as a teenager but in some ways i kind of wonder if everything in most organisms comes back down to those evolutionary priors and we're able to kind of play with um well even even a puppy is different to a full grown dog in terms of how they fit themselves into establishing their role in communication but we really do go outside of what our gut feeling tells us and start to sort of change the rules of the game so to speak um about causation and ontology and current utility so it's it's quite it's quite useful in terms of sort of generative discussion beyond like what this immediately says in a direct short-term interaction cool alex or yvonne any thoughts on that and and also just if you you can raise your hand if so um one other thought is the two by two kind of punnett square layout makes us think if we threw a dart it would be okay it's that's causation that's utility that's development that's evolution another spatial metaphor would be like a wave and a ripple on a wave so the big wave is evolution um and then there's smaller waves within that and so yes it totally builds on the evolutionary priors like you were just saying and it's totally about development because that never stops either and it's totally about the mechanism because without mechanism you don't have voice box or you don't have the audio processing so it's all of those and they're all happening in an overlapping way that's not quite reflected by the cleanliness of this two by two or even a continuum model cool any last thoughts on the table or can we get to the figures awesome so here we are in figure one and for for team com this was a uh really a great image that we had shared a lot before this discussion because it brings together the skeleton of active inference in a way which is this cycle that goes from internal states which influence through policy states action states action states influence external states of the world which are also influencing each other but we're not too worried about those external states influence our sensory states the retina and the tympanic membrane and then the sensory allow us to update our internal model of the world that's also generative so that's the very basic skeleton of the active inference cycle here reflected with just one agent in feedback with the world and so the the flow is always going this way but also it's happened all these things are happening all at once your legs are always engaged in an action state your retina is always involved in a sensory state even if your eyes are closed your internal states are always updating so they're all happening at once but we can kind of trace this causal diagram uh through time and then we also see represented in this image the equations and there is a level of math that is just sort of brought up in this paper and cited to other places for where people can learn more but suffice to say that each of these different uh sectors can be thought of as happening at various levels of mathematical description so in this center part of the image it shows like the the the minimal version of the uh math and then this is a little bit more fleshed out of an equation and then there are citations and future work to be done that really builds it out more fully so what does anyone else see in figure one i suppose not quite in figure one but in something you mentioned about the parts of your body that are doing the action they're doing the sensation um is a great skeleton like you said um but then there's all of the action that's happening constantly at sometimes consciousness at the subconscious level but likes the cods of the eyes so you're constantly having the small action to resample the environment or the like proprioceptive sensations of your legs so even within that larger cycle there's all of these much much smaller cycles happening where this perception action loop is functioning in like every part of the sensorium of your body very nice good point and this image has the sensations coming into the person and this kind of puts the black box or the markov blanket if you will around the whole person but it's a nested markov blanket situation so each neuron has internal states action states external states for the neuron which are other neurons other cell types and then sensations so this is something that is about the world and about people but it also exists as a skeleton whenever we want to draw the boundary around any kind of uh active inference system and thinking about communication more broadly than the human case is where we start to see oh yeah that's right um honest communication is the norm neurons don't expect each other to be deceptive sure it might happen but that's the decimal point and the real framework is about how cooperative communication goes down because adversarial relationships just don't persist over as long of time durations or spatial durations because they break down almost by definition and so thinking about this multi-level in space as well as multi-time scale kind of tying it back to the tim bergens levels um or four questions and there's a lot to this and it's awesome and one of the key contributions of the free energy principle which have to do with all of these equations that potentially despite the differences in the specifics between time scales and spatial scales and different modes of communication there may be some similar equations underlying the dynamics and that would be something quite fantastic so let's build on figure one to figure so here we go we have the same the same colors and the same layout here if you just block off one side of the picture and you just look at this side for example it would be the same thing the one person with a sensory uh the action states that influence the world i guess action and sense have been flipped here it's the action states that influence and then comes back through sense to the person but the two key differences here first off is instead of just internal states um it's now reflecting specifically the regime of attention so that's salience or attention and it's thought it could be thought of as a subset of the total internal generative model of the world by the brain but now we're talking about what specifically the in the internal model should attention be paid to and then the key here and this is almost the genesis of team com in some senses is the world for communicating agents is other communicating agents and it's a deceptively simple conclusion or claim because it's one thing when you have the person in feedback with the world that makes it it's kind of i'm thinking about tom hanks playing ping pong against a wall it's him he's he's in feedback it's a it's a dynamical system but the fixedness of his opponent makes the uh moves that he makes very very uh specified and easy to determine whereas once you even get two that are communicating and they have different attention they're paying attention different things because they have different internal models all of a sudden you get these totally novel and improvisational dynamics that they call thinking through other minds any thoughts on this that was a really great tom hanks ping pong metaphor though for i mean it'd be it would be that easy if you were just typing to a chat box and then it just copied and pasted exactly what you wrote back for example it wouldn't be quite a conversation and then you could imagine speaking to a chatbot or something like that and maybe attention is playing out differently on the computer versus in your brain so the internal states are different but again we kind of already knew that the action states are different the point is the shared epistemic resources so the epistemic being related to the knowledge the cues for the chat box what time the message was sent or who the person who sent it was the cues and the resources are a shared resource they're a shared niche and then each of those uh agents depending on which sense states make it to their regime of attention are going to update their action policies in a different way and so one other thing i really like about this framework for communication is it highlights the the necessity and the importance of alignment but not perfect alignment it actually as a starting place takes that different people are going to be coming from different perspectives it's not that there's one correct interpretation of what person a says and if person b gets it perfectly they succeed and if they don't get it perfectly they failed it's kind of an absolutist notion of and here we take a really fully pluralist approach to communication where you have two communicating agents they have different internal models of the and the communication is happening in this world that they share in the informational and in the physical niche that they share and it's actually an action-oriented interaction that the agents are engaged in that's happening as their very different internal models are aligning and so it highlights the process of alignment without being too hard on the need for a perfect that just in so many situations is a helpful starting point about where where am i at in this conversation where is my conversant in this conversation how are we going to align what is this alignment for any thoughts here uh let me try to end about maybe about actions because as for this model and everything is about alignment of the models and actions are like not side effects but not a primary one for example if a team or a company speaking about playing actions some kind of strategies or something like this basically it's again about alignments and that's why if a company or team want to develop something or make some it's need to planning recursively not from actions but from models and alignments yeah and to kind of give an example that if we were uh working together on a team to build a car the goal isn't to align on what the perfect car would be the goal is the action of actually making that car happen and that's going to require us to synchronize on mental states but also contribute our unique skills and perspectives and then to give one more example about action um declarative speech is an action it's an action it's a motor behavior it has to do with the way that muscles move so speech is an action this is one action that results in speed in sound clapping but their body actions that result in different cues and so placing speech uh fundamentally as an action state it helps us get away from this um well you have acts and then you have speech but of course there are speech acts and speech is an action and in pure conversation where the feedback is happening through conversation uh it gives us that same uh action-oriented uh model that we can apply to non-linguistic systems and now we're thinking about speech as an action in a communicating system so we can really build on the strengths of linguistics like jared was talking about without overly making it about the specifics of any human communication style or even a cultural communication style because speech is again just one of the many multi-faceted ways that we share information and that we call people's attention to different features of the world any thoughts on figure 2 or anything else any questions yeah stefan i think this is all this is also good for bringing um action into what language is about like action is out there in amongst the world as part of sort of interacting with rather than being these representations in the minds that codify everything so i think it's quite um it's quite useful to see that and see the world in the background um so yeah one thing i find in some ways i'll always be interested to see if there was a triad here to see how that would play out because because that might really bring that out um but it is interesting that the descriptions you were given there around how things are unfolding and i i didn't haven't quite unpacked all the equations but i think there's something useful in like there's something about the epistemic resources and more like general complex adaptive systems which maybe aren't entirely stable but kind of they're interacting and bounces off each other and the more internal stuff is much more noisy non-linear and it's a sort of achievement between like entities in the environment bouncing with each other and our kind of non-linear kind of free energy trying to understand what's that all about by minimizing free energy so i think this is actually quite there's quite a lot that can be played around awesome um yeah i agree about the the getting more than two actors so we can imagine another yellow box up here and through action states it's contributing to the epistemic resource and we can be super specific let's think about this video chat so the action states are what people are speaking about and the sensory states are happening to me through my eyes as i look at the screen and through my ears as i'm listening and through the sensations that i'm receiving i'm updating the regime of when i see a little uh cue an ostentive cue like a little blue box around the speaker that's putting my attention to not just where i should be looking but conceptually what i should be paying attention to um and then it also relates to the hermeneutics idea that jared was bringing up and so i'm imagining this epistemic resource our shared informational niche is we're all seeing basically the same six boxes on our screen our shared informational resources are identical and we can in a computer system even know that they're identical but each one of us are going to unpack in a different way what is being said and so it's like the book's on the table and someone says the meaning is in the book well it's not technically in the book it's not physically contained in the book that's a spatial metaphor that it's in the book and so where is the meaning well if the meaning is in the relationship between the regime of attention and the shared epistemic resources well then you get almost pluralism for free you get this idea that different people are going to by looking into the book in their own unique way they're going to unpack the book differently maybe one person doesn't read the language and so the book doesn't reduce their uncertainty about anything but for someone else who's very familiar with the work of the author potentially they can read between the lines and even see things or unpack things in the book that aren't directly there and so we get all these amazing special cases and and uh richness of coming out within this scaffold and so again just to go from figure one where okay this is just the minimal possible active inference cycle just external states internal states and then the markov blanket is being pierced on the inbound by sense and being pierced on the outbound by action let's just zoom one layer out now we're looking at the dyad and also we can just say the dyad's not just their um dissipating energy for nothing they're communicating to a line on we're here to reduce our uncertainty about this paper and about the free energy principle about active inference um as well as other goals and so all of this is already starting to come out just as we qualitatively look at it and i agree that the mathematics uh of complex adaptive systems and free energy principle provide a lot of those non-linear dynamics that we would want to see for a formal communicative framework i think it's a good time to show figure three this is the last figure in this sequence of three so here we're kind of just um specifying uh again just drawing out again this communication between two brains and it's one canonical loop of the coupled action perception cycle so we're keeping in mind it's always happening continuously and we're just going to trace out one loop it's kind of like we inject a little dye into the river we're just going to look at the trace as it goes just through one small part to get a bigger sense about the flow so it starts with my idea and my idea is going to influence my behavior that my behavior is going to fill out or pour into that epistemic niche my behavior in this case is a vocal behavior and then through our shared informational niche my behavior is going to result i mean this arrow you can imagine putting in other steps intermediating here like going through the ear and going through this cortex and that cortex but just broadly my behavior will result your version of my idea i said grab a gallon of milk and use you got it okay i got it a your version of my idea is going to uh result in combination this is the x your ideas and your version of my idea come together and that is going to influence your behavior then we're back right where we started where your behavior is going to say ah okay right they said exactly the thing that i wanted from the grocery store back to me so their behavior suggested that the attentional regime was there that the semantic content of what i wanted to convey was their i'm pretty convinced that my of what you thought i said was accurate aka the alignment was adequate doesn't need to be perfect but for the purposes of that communication the alignment was adequate and then my ideas and how i perceived your perception of them combined again in the generative model the brain and that's going to influence behavior i say okay great glad you got it here's five bucks go ahead and do this so this is just tracing one loop but already we get so many of these cool patterns and ways to think about how different sources of prior information like my idea your idea relate to the uh just in time information that we receive everyone has years and years and years of their own thinking and then someone says hey what do you think about this policy and they're combining this recent sensory information from the person asking the question with a really big and rich sensory regenerative model of the world and so we we can capture that short term information as well as these deeper developmental and evolutionary aspects of thought together in this type of a model and again it's just a wireframe but what does this um lead anyone to think about or what what might be uh uncertainty that some of our listeners would have about this figure one two three sequence yeah shannon i suppose in this figure one two three sequence um everything seems like this event right now um and what's left out is this iterative process um so like going through the these chains of language learners and as you get to the end you might get a simple construction by the time you get to the end or chains of interactions between people where my idea is continuously being just slightly adjusted by your version of my idea which slightly adjusts each other node in this figure until we as a dyad um sort of start to align on the like sort of final state or something of our interaction um which is brought up a lot in the paper but it's not it's not super clear just from seeing these these three figures sort of in a sequence yep anyone else there maybe sasha how do you think this relates to improvisation or where improvisation takes interactors um yeah i i quite like this figure um adding the extra or kind of the next logical step that um it's about not just how i'm expressing my idea but what is your version of the idea that i'm expressing um and yeah i think it's um yeah very relevant for uh improvisation and i'm just thinking of like context uh specific clues that we get from iterative interactions and um i'm thinking like almost like a secret language you know that you'd be able to speak in code once you did it once to the person you would know what these words mean kind of going forward but um as shannon mentioned like that wouldn't be captured in in this um in this figure uh precisely um or explicitly um but to know that uh when you're in this location you call something by a secret code name um these are the kinds of things that are really important for communication that wouldn't make sense in another context or place in your life cool one other thought and i wouldn't quite characterize active inference as an optimistic theory i i don't know if it's optimistic or not but what i really like as far as a starting point for our uh often discordant world is it starts from a notion that everybody has a unique perspective and through shared resources they're coming into alignment and that's why we need intercultural non-violent communication all these types of ideas whereas the other way to think about would be somebody has the true idea of what this ism is you know socialism is x and if this person has the true one or this book has the true one and then those who diverge from it it's an error term and so instead of thinking about an absolutely correct idea and then divergence that different individuals have whether a person is the one who holds the absolute truth or whether no one has the absolute truth and everybody's deep in the error term we can just think about meeting people where they are starting with the states that people actually have instead of um starting from some hypothesized perfect state and then trying to bring people into alignment there i think it's just a very helpful idea and the secret language and the intermodal to kind of put one more finish on that if somebody puts their um scare quotes on the video uh with their fingers which is a non-verbal cue that suggests i'm using this term ironically or it could mean different things but importantly it might mean different things in different cultures so one might want to be very careful when they're using non-verbal or secret or culturally specific cues because if someone's not looking at your video square the second that you make those scare quotes or you lag out for a second in the context of online communication your semantic content will not update their model appropriately they may think that you're saying something that's quite the opposite of what you intended to say and so a lot of the times when people are talking about how to improve the communication throughput in online meetings and there's distraction that's kind of the opposite of attention there's all these issues that confront online meetings and online teams and people will say well use the body language and use nonverbal cues and yeah it's important to use those to the extent you can but also within this framework we can already start to think about some cases where maybe that wouldn't help as much as we thought especially when we're in the challenging communication contexts like low bandwidth or different language speakers so these are the communication uh scenarios where we actually want to have useful theory we don't just want to describe the easy low-hanging fruit of two people who are twins and speak the same language and read the same books we want to talk about the professor who's speaking with a child about math and we want to get them together on the same vocabulary and cut those are the communication scenarios that really are meaningful to explore and the ones that are so natural to this skeleton but are not as easy to model in other frameworks um any other thoughts on this figure yeah stefan yeah i think it's like you just mentioned in fact that they say canonical like this feels like it makes more sense for deductive thinking that you're trying to line up because just having the brain there obviously that kind of sort of it's it's in this kind of slightly disembodied slight way and so okay i have a deducted idea and then he was like okay what is that and this this could make sense now if we're gonna ground it more in like my state or my embodied um feeling about something more generally if i'm working in a more iterative or inductive way or trying to sort of feel the situation abductively i don't know if this probably would be quite how i would maybe one of the other diagrams is more appropriate but this may be that idea that an idea pops out which would probably be a deontic or you know deductive here's a new equation and now it's like tick tick tick tick tick you can imagine it being a bit more schematic like this nice agreed and um the five e's uh that are sometimes brought up in these papers inactive embodied and cultured all these ways that go beyond just idea teleporting into another idea that is uh it's not even like that's the payload of communication it's just we can come at it from the other side by thinking okay what are the bodies doing what is actually being perceived in that embodied setting and then yeah the ideas are there but um we could just as well have had stick figures instead of two brains because or we could have had a stick figure over here for my behavior or mouth or something so let's go to figure four and five closer to empirical uh figures and these are specifically in the context of birdsong so here's the upper part of figure four and what is happening here anyone wanna just describe what they see when they look at this well on the x-axis of these diagrams yeah sasha go ahead um yeah so this is looking at um birdsong improvisation and um sound is a really nice way to do that because you can have a kind of a dimensional reduction and look at just the outgoing sounds um instead of trying to analyze like body posture and interaction which is um quite challenging and um when uh birds sing together versus sing singing alone um we might imagine that um we could take the two of them singing alone and average it out or something like that that could be um one possible outcome of when we put them together um and then the other outcomes are that they improvise and create something completely different that isn't just the average of the two agents yes so from the caption it says two birds endowed with prior expectations about the hidden states generating a shared birdsong narrative and i love that they called the sheriff bird song narrative um the way that it works it says sing for two seconds and then listen for a response so that's why on the x-axis here where we have time in seconds you have one two in blue and then listen for response one two listen for a response the posterior expectations so that's after the priors have been updated what you get are posteriors the posterior expectations for the first bird are shown in red and expectations for the second bird are shown in blue both as a function of time the left panel shows the chaotic and uncoupled dynamics when the birds cannot hear each other while the right panel shows the synchrony in the hidden states that emerges when the birds exchange sensory signals so the the left image is kind of like two people they're in two recording studios that are separated they can't hear each other and they're both singing a song it's a call and response song and under the uh model or the ecosystem where they're not sharing a shared informational niche they're in two isolated recording studios there would be no expectation that the turn taking would happen um if if someone took off their headphones and just was talking randomly there'd be no expectation of turn taking in fact to be quite impossible and so in this singing alone like singing alone with two of the birds who are both seeing to themself um you see that there's basically uh there's there's strong discordances where the one bird is like okay i'm gonna sing now and the other one's like i'm gonna sing now and also just um i think this is more of a schematic but let's just think about this as like a prefrontal cortex this is where the higher order thought is happening and then this is like a motor cortex this is where the speech is being generated between um the more basal regions of the brain and the more cognitive elements so on this left side we have the bird is singing it's it's um infer it's deciding when to sing and also listening for a response but it's not hearing any response so it's uncoupled you get uncoordinated dynamics however when you allow for the birds to sing together you see that they can synchronize so that they can actually um communicate and really importantly it's not enough just to have the two singers in the same recording studio or have them listening to the same sounds they actually have to have priors that they want to strive towards alignment because if someone says yeah yeah yeah this is a turn-taking song but i don't care about that well then you're also going to get the emergence of uncoordinated behavior so you need the sensory coupling which takes place through the shared informational niche and the adaptive prior that the person or the bird that you're communicating with is an agent like you who also wants to synchronize because that's the cooperative part if you're prior about your your um conversions is that they don't want to synchronize or that they're not going to synchronize well then you might not synchronize with them you might not give them the opening to synchronize so when we start from this place of communicative communicate or cooperative communication that means we're going to need that sense coming to the person appropriately so that they can turn take but then at an internal generative model level we also want to have this deep prior about our communication that we want to synchronize and so a lot of the later work after this paper has gone into this more about how individuals who just want to believe or want to reduce uncertainty about a higher order entity that they belong to that is what can actually lead to group formation so this is a really interesting uh concept any thoughts on the top half of figure four or can we jump to the bottom of figure four cool so here we are on the bottom of and on these plots though the x-axis is the second level expectations which is like what you expect to be hearing the first level i believe is referring to what um like i'm my first level expectations of my speech are just what i'm generating whereas the second level expectations is what do they think um what do i think that my conversant is getting and so on the x-axis of both of these plots are the second level expectations of the first bird and the y-axis is the second level expectations of the second bird so here what's being communicated is a number between negative 20 and 60. but it's just a number line so yeah how how many uh or what's the price of oil today negative 20 to 60. we're trying to reduce our uncertainty about this shared niche and what we're seeing is um we can just go to the right side first the y equals x line this one to one line would mean the second level expectations of the first bird is negative 10 and the second bird got negative 10. so this line represents points on this line represent where there's an alignment of because the expectations of the first bird are aligned with the expectations of the second bird and we can see it really clearly on this right side when they're synchronized that whether the value being communicated is negative 10 or whether it's 50 or 60 it's a dynamical system so we're still getting a trace that kind of flows above and beyond the line but we're spending most of our time on alignment about what is being said but if we go back to this isolated case where there's no sensory coupling and there's no ability for even if there were a deep prior that said i want to be aligned with another bird we're spending as much time off of this y equals x line as we are spending near it because the birds are just uncoordinated in their singing and so this is just showing this is what happens when you don't have alignment and this is what happens when you have synchronization and a manifold is a lower dimensional line let's just go with for now that is a dimensional compression from a higher dimension so the points are in very higher dimensional space but when you put them down on the two-dimensional piece of paper it makes a parabola or in this case it's a higher dimensional space of two dimensions the expectations of the two birds yet it exists or it at least converges towards this one-dimensional manifold which is just a line representing from negative 20 to 60 what the actual communicated state is so this is just another a phase space dynamical representation of what alignment looks like in the uncoordinated case and then in the coordinated case where you have sensory coupling and those deep priors for alignment any thoughts on this one cool let's just go to five so that we can finish the figures and have any last thoughts okay anyone want to take a first stab on figure five what is being communicated here so in the caption of the figure it writes a shows changes in the posterior expectations of an order parameter for the first bird in blue and a second bird in green determining the chaotic structure of the songs depicted in figure four and so and it's as a function of the number of exchanges so here the x-axis isn't the time in seconds with the two seconds of singing two seconds of listening now we're gonna those that two seconds of singing and listening we're going to condense that into one exchange so now we're kind of thinking in that iterated um way so we're starting to get towards iterated communication here by condensing through time and almost looking at kairos rather than kronos and then we have this uh value of a of a order parameter and uh yes so we have an order parameter and what we see is that the order parameter starts off high again the green and the blue are the two birds and the green bird is starting starts higher organization in both of these traces the more dim one the learning alone which is these two dimmer traces and this the line represents the the mean of these simulations and the the shaded area represents the um 90 bayesian confidence intervals just to clarify that so we can start up here this green bird starts at perfectly ordered and then the blue bird 0.5 so a little bit more disorder maybe this is like a mentoring relationship between a more encultured bird and a younger bird who's on to genetically learning how to communicate and what we see is that when we have the ability to learn together that the order parameter stays high and we can actually increase the order of disordered agents so for example if five of us had a really good internet connection and one person was lagging a lot through cues we could help order the person's communication even if they had lack however if we learn alone then the order parameter even though it starts very high for the green bird and a medium for the bluebird in both cases you get this exponential decay so that after 30 bouts so 30 times 4 seconds 30 exchanges 30 choruses you have zero correlation the order parameter is zero because they started out like partially synchronized just by happenstance like you told the two singers in the two different recording studios okay start singing and initially they started just by coincidence synchronized but rapidly that decays and it decays monotonically there's no chance they're ever going to recoup alignment because they're not coordinated but again if you are coordinated you can learn together and just like sasha was saying um when you have improvisation and feedback it's not just that the green parameter stays at one and the blue one converges toward it it's actually that the two converge together and they meet in the middle which is just so um reflective of communication and we're getting this just from birdsong this isn't at the semantic level this isn't at any other higher order level sasha um this also makes me think of um like you know the two birds uh teaching each other the the song or coming to a kind of more ordered state and mentoring relationships as well as teams um and while i think um people kind of intuitively understand that there's going to be um some sort of meeting in the middle linguistically to describe it um but how to rise to that highest order um and how to converge on something that more ordered and not drop down to something that is less ordered um i think is an interesting question for teens and maybe is a little more straightforward in like a teaching relationship where um one person is quite fixed and um both people know that that's the fixed point to rise to um from the student's perspective yep and it's kind of like when we say everybody can learn from everyone even if one person's knowledge base totally encompasses someone else's knowledge base at the very least we're learning how to communicate we're learning how to learn so really we can honestly say everyone can learn from every interaction with every and now we have a framework to really go into that let's just close it out with five b and c so this is reminiscent of figure four where again we have the expectations um here are the second level and uh expectations of the first bird and then the expectations of that second bird um and again we have this y equals x manifold that would reflect points on the line uh being in perfect attunement where the first bird is expecting to communicate 40 and the second bird's expectation is also 40. so that's the uh alignment of the internal models through coupled communication and a deep prior that you want to align and what we see is that before learning so now we're going to be talking about this learning together trace so before learning is here so before learning it's not as wacky as figure four was where these are like all over the place they're it's truly uncorrelated it's like a cloud of points here we're starting with a partial ordering the green order parameters one the blue or the parameters point five so this um oval is halfway between perfect and random um and there's some math that we could go into but not today but it starts off partially organized and you need to kind of start from a place of partial organization for communication and that's actually granted by evolution and the synchronization manifold before learning gets tightened to the synchronization after learning where the parameters have been improved so that now the birds are in improved alignment yeah stefan is your hand raised yeah sorry i was just trying to yeah yeah i just thought one thing was interesting on the other is the more experienced birds um it's like there's still a slight dip in their coordination um as they learn as they as you know in their kind of um now maybe there's a trade-off but the green one still goes down a bit and is there maybe a sense that you need to relax a little bit of the rigidity of the pattern to enable some compromise or some sort i'm not exactly sure how they came up with the the way that that that model works but that i just noticed that i thought that's kind of interesting there that there's a slip in that and maybe that's just a degradation of degradation over time it just happens but there seems to be something about that ability to relax the rigidity of the model that you use totally agreed and i'm i'm thinking of a math setting where the equations are in the textbook and then one person might be very familiar with them one person has less familiarity so the green person is teaching math with the blue person and it's not teaching two that's that absolutist idea they're in a shared alignment they want to align around a proper understanding of math but the teacher person still has to be ready to address unexpected questions from a student which may be extremely valid and actually push the field forward by leaps and bounds by bringing that beginner's mind to the equations um and if nothing else the teacher might have to resort to some informal metaphors or say well this this parameter wants to do this that's kind of if you just think about it from the textbooks perspective it reflects a divergence of that order parameter but it's being used towards this higher end which is the shared alignment on the concepts and so yeah very interesting and i i wonder um about that too um and maybe that's reflected by initially the blue line takes a larger excursion away from order but then it spends it then converges a little bit more so that's like when you have a first conversation with somebody and it's like um so where are we coming from you're taking more of an excursion towards this order to be more in the explore phase and then once you have a better sense about where the person's coming from you can enter into a more exploit type phase um i think that's um almost all for our time and that's all the figures of the so on a closing page just if anyone wanted to share um uh closing thought or if they wanted to share a question that they had an idea or a future experiment so anyone feel free to jump in or raise your hand uh i just want to say an idea what you should never sing alone is it a russian proverb yes well singing alone could be nice i mean solo performance can be beautiful and uh it makes me think about seeing the national anthem which is often done with a solo performance they might do a little bit of more vocal flair or they might draw out a certain thing that would totally disrupt the um if multiple people were trying to sing it but when one person is doing it they can have a lot more artistic license but if you're in a choir context where it has to be synchronized you you can't just say well today i'm going to sing it this way it's not going to sound good um and so there's there's there's maybe times to sing alone but it's not singing together to sing alone that's for sure but are they singing alone i mean if you're performing you're singing performing for so you're engaging with an audience of some sort or a recording studio which still has a implied audience very true and um they're the the um they have a paper this figure five caption it's a duet for one and so i think that the joke there is it's one dyad it's it's a conversation for one this is a conversation for one shared markov blanket it's a conversation for one informational niche team com it's one conversation that's happening it's not six conversations but there are six perspectives happening and so that's where i find active inference to be such a powerful framework because it helps us um really give full value and cognition to six unique perspectives coming together to one shared niche and so one fallacy would be well six shared perspectives means there's no way we're gonna synchronize that's a little bit too defeatist and then the other alternative would be well because there's one informational niche there's only one conversation and if you don't get it then you're not with a picture and that's also not right but somewhere in the middle we can have a shared niche and our own perspectives sash um yeah just to close it out with the the meta uh connection um you know with the words teaching each other how to sing and coming to a higher understanding um we're trying to come to a higher understanding of active inference and it was really great to have the first author of the paper here to kind of align us and set the order so that was really nice yep yep he provided context to what we discussed and so it was very very nice to have jared come through any other thoughts great well that concludes um team com podcast number three i really appreciate all of our participants as well as all of our listeners and we are interested in staying engaged on these topics and hearing people's feedback we want to be really uh inclusive and engaging with how we're talking about these topics from conceptual perspectives uh and more so please be in touch with and we will have this every two weeks so we really look forward to participating with many of you soon and i'm going to terminate this live stream thanks again everybody for participating and listening thank you